Sheth L.U.J and Sir M.V College

Aim: *Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.*

Input:

```
# Diffie-Hellman Key Exchange Algorithm
print("----- DIFFIE-HELLMAN KEY EXCHANGE -----")
# Publicly known values
p = 23       # Prime number
g = 5        # Primitive root of p

print("\nPublicly known values:")
print("Prime number (p):", p)
print("Primitive root (g):", g)
# Alice's private key
a = int(input("\nEnter Alice's private key: "))
# Bob's private key
b = int(input("Enter Bob's private key: "))
# Calculate public keys
A = pow(g, a, p)
B = pow(g, b, p)

print("\n----- PUBLIC KEY EXCHANGE -----")
print("Alice's public key:", A)
print("Bob's public key:", B)
# Calculate shared secret key
alice_shared_key = pow(B, a, p)
bob_shared_key = pow(A, b, p)

print("\n----- SHARED SECRET KEY -----")
print("Alice's shared secret key:", alice_shared_key)
print("Bob's shared secret key:", bob_shared_key)
# Verify that both keys are equal
if alice_shared_key == bob_shared_key:
    print("\nKey Exchange Successful!")
    print("Both Alice and Bob have the Same Shared Secret Key.")
else:
    print("\nKey Exchange Failed!")
```

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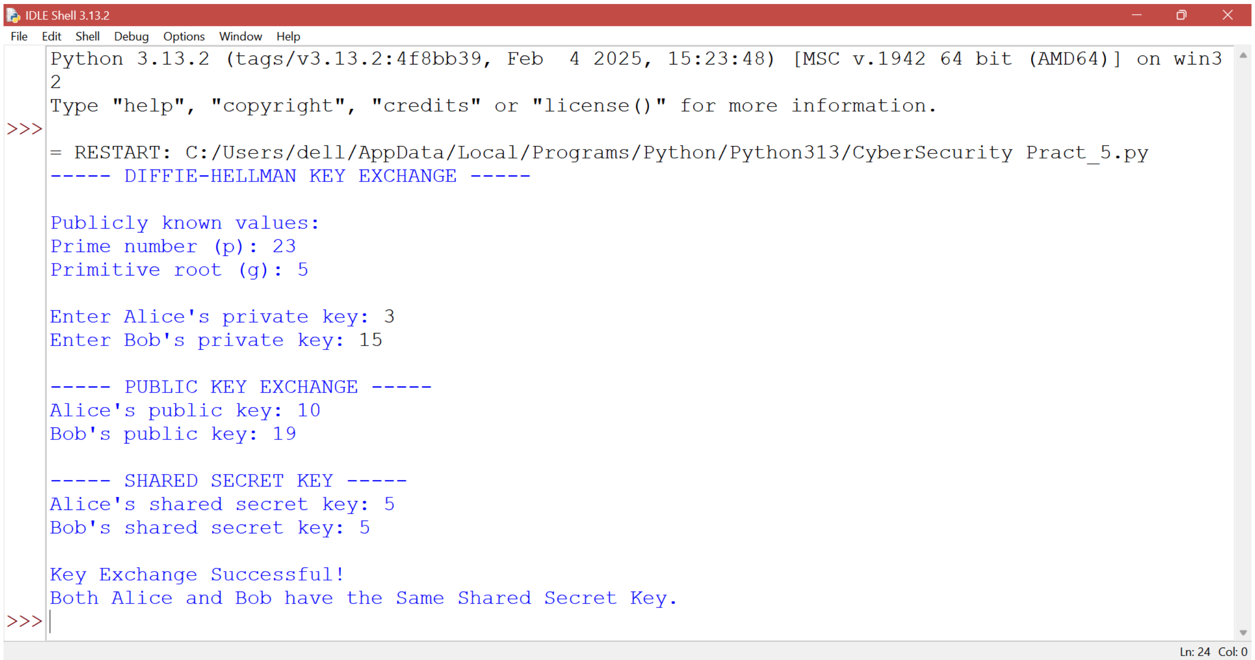
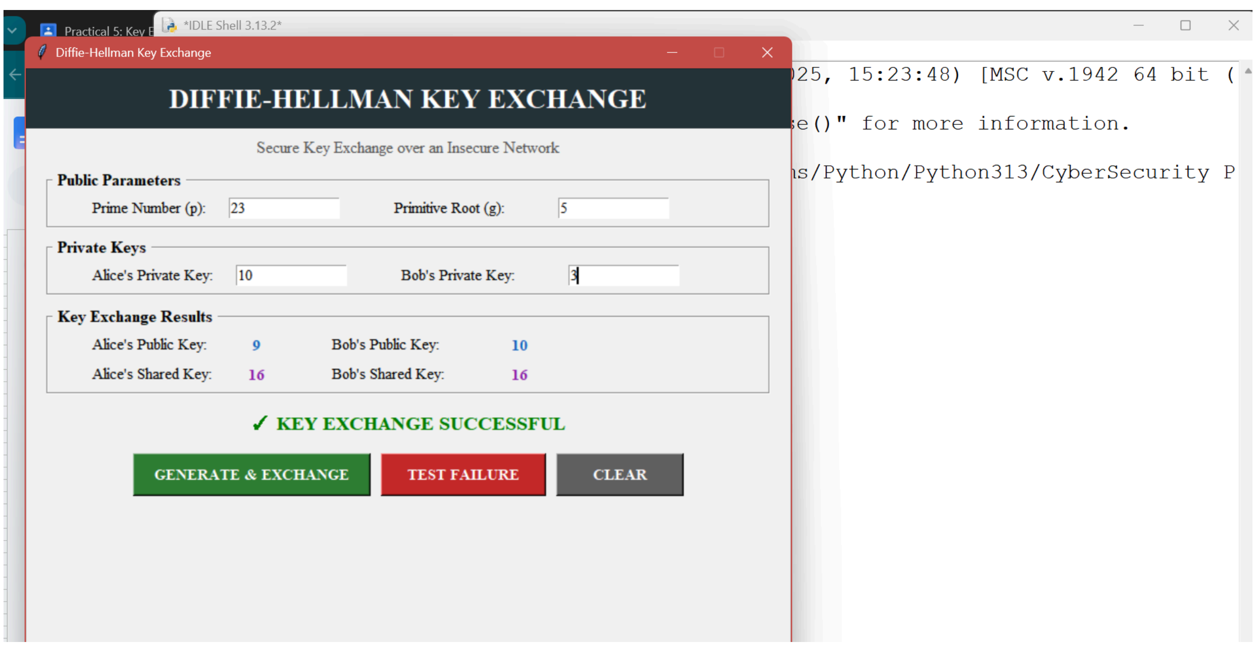
Output:



GUI



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